Review Questions for Lectures 9 and 10

1. In the game Wheel of Fortune, a contestant spins a wheel, and when it finishes spinning, it will point to one of six prizes, all of which are equally likely. Five of the prizes are small, but one of them is one million dollars. Suppose you play the game and you are only allowed to spin the wheel once. What is the chance you win the million dollars?

2. Now suppose you are given two options in the Wheel of Fortune game above:

   a. Spin the wheel once, and claim the prize it lands on. Spin the wheel again, and claim that prize as well if you didn't already win it on your first spin.
   b. Spin the wheel once, and claim the prize it lands on. Move the wheel one spot forward, and claim that prize as well.

In each case, what is the chance you win the one million dollars?

3. Repeat the previous two questions, but this time assume that there are two spots on the wheel with a million dollar prize, instead of just one. Does it matter which spots have the million dollars?

4. A box contains 6 tickets with labeled top and bottom sides. One of the tickets is black on both sides. Two of the tickets are white on both sides. The three remaining tickets are black on the bottom side and white on the top side. A friend pulls a ticket out of the box at random and places it on a table.

   a. What is the chance that the top side of the ticket is white?
   b. Given that the top side of the ticket is white, what is the chance that the bottom side of the ticket is black?

5. If A is "rain on Saturday," and B is "rain on Sunday," and $P(A) = 0.3$ and $P(B) = 0.6$,

   a. What is the maximum possible value of the probability of rain on both Saturday and Sunday?
   b. What is the minimum possible value of the probability of rain on both Saturday and Sunday?

6. If A is "rain on Saturday," and B is "rain on Sunday," and $P(A) = 0.4$ and $P(B) = 0.8$,

   a. What is the maximum possible value of the probability of rain on both Saturday and Sunday?
   b. What is the minimum possible value of the probability of rain on both Saturday and Sunday?

7. Consider three events A, B, and C. Suppose that $P(A) = 0.2$ and $P(B) = 0.5$ and $P(C) = 0.7$. What is:

a. the maximum possible value of $P(A$ and B and $C)$,
b. the minimum possible value of $P(A$ and B and $C)$,
c. the maximum possible value of $P(A$ or B or $C)$, and
d. the minimum possible value of $P(A$ or B or $C)$.

8. Consider three events A, B, and C. Suppose that $P(A) = 0.8$ and $P(B) = 0.9$ and $P(C) = 0.95$. What is:

a. the maximum possible value of $P(A$ and B and $C)$,
b. the minimum possible value of $P(A$ and B and $C)$,
c. the maximum possible value of $P(A$ or B or $C)$, and
d. the minimum possible value of $P(A$ or B or $C)$

9. Suppose you draw two cards from a well-shuffled deck. Let A be the event that the first card is red, and let B be the event that the second card is red. Explain in your own words the frequency interpretation of each of the following two statements:

a. P(A | B) = 25/51
b. P(B | A) = 25/51

10. If A is "rain on Saturday," and B is "rain on Sunday," and $P(A) = 0.4$ and $P(B) = 0.4$, then

a. Tell a story where $P(A$ and $B) = 0$.
b. Tell a story where $P(A$ and $B) = 0.4$.